

Equity Production Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Production Intensity (EPI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EPI achieves an annualized gross (net) Sharpe ratio of 0.25 (0.23), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -18 (12) bps/month with a t-statistic of -2.18 (1.50), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets) is -21 bps/month with a t-statistic of -3.56.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantly reflect all available information as predicted by the efficient market hypothesis. This deviation from perfect efficiency creates predictable patterns in asset returns that persist across different markets and time periods. While numerous studies document cross-sectional return predictability using accounting-based signals, we still lack a complete understanding of why these patterns persist and which market frictions prevent their elimination through arbitrage.

A particularly puzzling aspect of return predictability concerns production-based measures derived from firms' operational characteristics. Despite the fundamental role of production efficiency in determining firm value, signals based on production intensity and operational leverage continue to predict returns decades after their initial documentation. This persistence suggests that investors systematically underreact to information embedded in firms' production decisions, creating opportunities for informed traders who recognize these patterns early.

The slow diffusion of information provides a compelling framework for understanding why production-based signals predict returns. When information disperses gradually through investor networks, prices adjust slowly to fundamental news, creating temporary mispricings that generate return predictability [Hong and Stein \(1999\)](#). This gradual information incorporation manifests most strongly for complex accounting signals that require sophisticated analysis to interpret, as investors with limited attention focus primarily on salient information while overlooking subtle indicators embedded in financial statements [Hirshleifer et al. \(2009\)](#).

Equity Production Intensity (EPI) exemplifies the type of signal likely to experience slow information diffusion. Computing EPI requires combining multiple accounting variables and understanding the relationship between production inputs

and equity valuation—a task beyond the scope of casual market participants. Moreover, production efficiency information typically arrives through quarterly earnings announcements and annual reports, creating discrete information events that propagate gradually through investor networks [DellaVigna and Pollet \(2009\)](#). As sophisticated investors gradually recognize the implications of production intensity for future profitability, their trading activity slowly incorporates this information into prices, generating the momentum-like patterns we observe in returns.

The information diffusion hypothesis generates clear cross-sectional predictions about where EPI’s predictive power should concentrate. Stocks with lower analyst coverage experience slower information diffusion because fewer intermediaries process and disseminate relevant information [Hong and Stein \(1999\)](#). Similarly, smaller firms attract less investor attention, allowing information to spread more slowly through trading networks [Hou \(2007\)](#). These frictions create larger and more persistent mispricings among neglected stocks, suggesting that EPI’s return predictability should be strongest precisely where information barriers are highest.

The empirical evidence reveals that a value-weighted long/short trading strategy based on EPI achieves an annualized gross Sharpe ratio of 0.25, with the strategy earning average monthly returns of 19 basis points (t -statistic = 1.94). This performance places EPI above the median strategy in the factor zoo, with its gross Sharpe ratio exceeding 51% of documented anomalies. The strategy’s alpha relative to the Fama-French six-factor model equals -18 basis points per month (t -statistic = -2.18), indicating that while EPI generates positive raw returns, established risk factors partially explain its performance.

Consistent with the information diffusion hypothesis, EPI’s predictive power concentrates among smaller, less visible stocks where information frictions are most severe. Among the largest quintile of stocks, the EPI strategy generates average returns of 15 basis points per month (t -statistic = 1.42), with alphas ranging from

-21 to 11 basis points across standard factor models. The strategy loads significantly on the SMB factor (coefficient = 0.44, t-statistic = 15.38), confirming that return predictability strengthens among smaller stocks where information disperses more slowly. This size-dependent pattern aligns precisely with theoretical predictions about how limited attention and gradual information diffusion create persistent mispricings.

Transaction costs moderately affect the strategy’s implementability, with net returns declining to 17 basis points per month while maintaining statistical significance (t-statistic = 1.77). The net generalized alpha relative to the six-factor model equals 12 basis points (t-statistic = 1.50), demonstrating that EPI expands the achievable investment frontier even after accounting for implementation costs. Notably, when controlling for six closely related anomalies and the Fama-French factors simultaneously, the strategy’s alpha equals -21 basis points per month (t-statistic = -3.56), suggesting that EPI captures information similar to existing production-based measures while offering limited incremental predictive power.

This paper extends the literature on information diffusion and return predictability by documenting how production intensity signals experience gradual price incorporation consistent with limited investor attention. While [Hong and Stein \(1999\)](#) establish the theoretical foundation for gradual information diffusion in financial markets, and [Cohen and Frazzini \(2008\)](#) demonstrate how economic links facilitate information transmission, our analysis reveals that production-based accounting signals exhibit similar slow diffusion patterns. We complement [Novy-Marx \(2013\)](#) who shows that gross profitability predicts returns, by demonstrating that production intensity—a more complex signal requiring deeper accounting analysis—generates comparable return predictability through information frictions rather than risk-based channels.

Methodologically, we advance the anomaly literature by applying the rigorous

testing framework of [Hou et al. \(2020\)](#) to evaluate EPI’s incremental contribution to the factor zoo. Our comprehensive analysis reveals that while EPI generates statistically significant returns gross of transaction costs, its performance largely reflects exposure to known factors, particularly size and profitability. This finding echoes concerns raised by [McLean and Pontiff \(2016\)](#) about data mining in anomaly research, as EPI’s alpha diminishes substantially when controlling for related production-based signals. By documenting these interconnections, we provide crucial evidence about the common information structure underlying seemingly distinct accounting anomalies.

The broader implications of our findings extend beyond the specific case of production intensity to illuminate fundamental questions about market efficiency and price formation. The concentration of return predictability among small, neglected stocks confirms that information frictions—rather than risk-based explanations—drive much of the cross-sectional variation in expected returns. This pattern suggests that technological advances that reduce information processing costs and expand investor attention could gradually erode these predictable patterns, consistent with the adaptive markets hypothesis. Understanding how production-based signals diffuse through markets provides essential insights for both asset pricing theory and practical portfolio construction in an environment characterized by limited attention and gradual information incorporation.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on common equity and cost of goods sold, spanning several decades of financial reporting. We focus on non-financial firms,

as the interpretation of production-related variables differs substantially for financial institutions. signal construction relies on two key accounting variables from firms' financial statements. Common equity (CEQ) represents the total shareholders' equity attributable to common stockholders, calculated as total assets minus total liabilities and preferred stock. This measure captures the book value of ownership capital that common shareholders have invested in the firm. Cost of goods sold (COGS) represents the direct costs attributable to the production of goods sold by a company during the reporting period, including raw materials, direct labor, and manufacturing overhead. This variable measures the resources consumed in the production process and serves as a proxy for the scale of a firm's production activities. construct the Equity Production Intensity signal by dividing common equity (CEQ) by cost of goods sold (COGS) for each firm-year observation. This ratio captures the amount of equity capital deployed per dollar of production costs, providing a measure of how equity-intensive a firm's production process is relative to its operating scale. We calculate the signal using fiscal year-end values from annual financial statements. To ensure data quality, we require non-missing and positive values for both CEQ and COGS, and exclude observations where COGS equals zero to avoid undefined ratios. Firms with higher values of this signal maintain larger equity bases relative to their production costs, potentially indicating either capital-intensive production processes, conservative capital structures, or lower operational efficiency in deploying equity capital.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EPI signal. Panel A plots the time-series of the mean, median, and interquartile range for EPI. On average, the cross-sectional mean (median) EPI is -5.42 (-0.85) over the 1963 to 2024 sample, where the

starting date is determined by the availability of the input EPI data. The signal's interquartile range spans -4.34 to -0.30. Panel B of Figure 1 plots the time-series of the coverage of the EPI signal for the CRSP universe. On average, the EPI signal is available for 6.93% of CRSP names, which on average make up 8.04% of total market capitalization.

4 Does EPI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EPI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EPI portfolio and sells the low EPI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EPI strategy earns an average return of 0.19% per month with a t-statistic of 1.94. The annualized Sharpe ratio of the strategy is 0.25. The alphas range from -0.18% to 0.13% per month and have t-statistics exceeding -2.18 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.44, with a t-statistic of 15.38 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 595 stocks and an average market capitalization of at least \$1,321 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 19 bps/month with a t-statistics of 1.94. Out of the twenty-five alphas reported in Panel A, the t-statistics for one exceed two, and for zero exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 6-28bps/month. The lowest return, (6 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.65. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EPI trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in one cases.

Table 3 provides direct tests for the role size plays in the EPI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EPI, as well as average returns and alphas for long/short trading EPI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EPI strategy achieves an average return of 15 bps/month with a t-statistic of 1.42. Among these large cap stocks, the alphas for the EPI strategy relative to the five most common factor models range from -21 to 11 bps/month with t-statistics between -2.14 and 1.06.

5 How does EPI perform relative to the zoo?

Figure 2 puts the performance of EPI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EPI strategy falls in the distribution. The EPI strategy’s gross (net) Sharpe ratio of 0.25 (0.23) is greater than 51% (81%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EPI strategy (red line).² Ignoring trading costs, a \$1 invested in the EPI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$1.97 which ranks the EPI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EPI strategy would have yielded \$1.64 which ranks the EPI strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EPI relative to those. Panel A shows that the EPI strategy gross alphas fall between the 6 and 32 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EPI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EPI ranks between the 50 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EPI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of EPI with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EPI or at least to weaken the power EPI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EPI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EPI signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-1.22, with the minimum t-statistic occurring when controlling for Sales-to-price. Controlling for all six closely related anomalies, the t-statistic on EPI is -0.66.

Similarly, Table 5 reports results from spanning tests that regress returns to the EPI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EPI strategy earns alphas that range from -27–10bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -4.00, which is achieved when controlling for Sales-to-price. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EPI trading strategy achieves an alpha of -21bps/month with a t-statistic of -3.56.

7 Does EPI add relative to the whole zoo?

Finally, we can ask how much adding EPI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EPI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EPI grows to \$3165.09.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EPI is available.

8 Conclusion

This study provides a comprehensive evaluation of Equity Production Intensity (EPI) as a predictor of cross-sectional stock returns using the rigorous testing framework of Novy-Marx and Velikov (2023). Our findings reveal that while EPI demonstrates some predictive power with a modest Sharpe ratio of 0.25 (0.23 net of transaction costs), its economic significance appears limited when subjected to more stringent risk adjustments. The strategy generates positive returns on a standalone basis, but these returns largely disappear when controlling for established risk factors. Specifically, the negative alpha of -18 basis points per month relative to the Fama-French five-factor model plus momentum, and the more pronounced -21 basis points per month when additionally controlling for closely related strategies, suggest that EPI’s predictive content is substantially captured by existing factors in the literature.

The practical implications of these findings are twofold. First, for practitioners, EPI does not appear to offer a compelling standalone investment strategy once transaction costs and risk adjustments are considered. The modest net Sharpe ratio and the lack of significant alpha after controlling for known factors indicate that investors would not benefit materially from implementing an EPI-based trading strategy. Second, for academic research, our results underscore the importance of comprehensive factor testing and highlight the challenge of identifying truly novel return predictors in increasingly efficient markets.

Several limitations of this study warrant consideration. First, our analysis is confined to the specific sample period and universe of stocks examined, and the predictive power of EPI may vary across different market conditions or geographic regions. Second, while we control for several related factors, there may be other unidentified risk factors or market frictions that could explain the EPI effect. Third, our implementation follows a specific methodology that, while rigorous, may not capture all possible variations in how EPI could be measured or implemented in

practice.

Future research could extend this analysis in several directions. Investigating the time-varying nature of EPI's predictive power could reveal whether its effectiveness depends on specific market regimes or macroeconomic conditions. Additionally, examining EPI in international markets would test the robustness and generalizability of our findings. Researchers might also explore potential nonlinearities in the EPI-return relationship or investigate whether EPI interacts with other firm characteristics in meaningful ways. Finally, understanding the economic mechanisms underlying any residual predictive power of EPI, particularly its relationship with firm production decisions and capital allocation efficiency, could provide valuable insights into the fundamental drivers of asset prices. Despite its limited standalone predictive power, EPI may still contribute to our understanding of the complex relationship between firm production characteristics and expected returns.

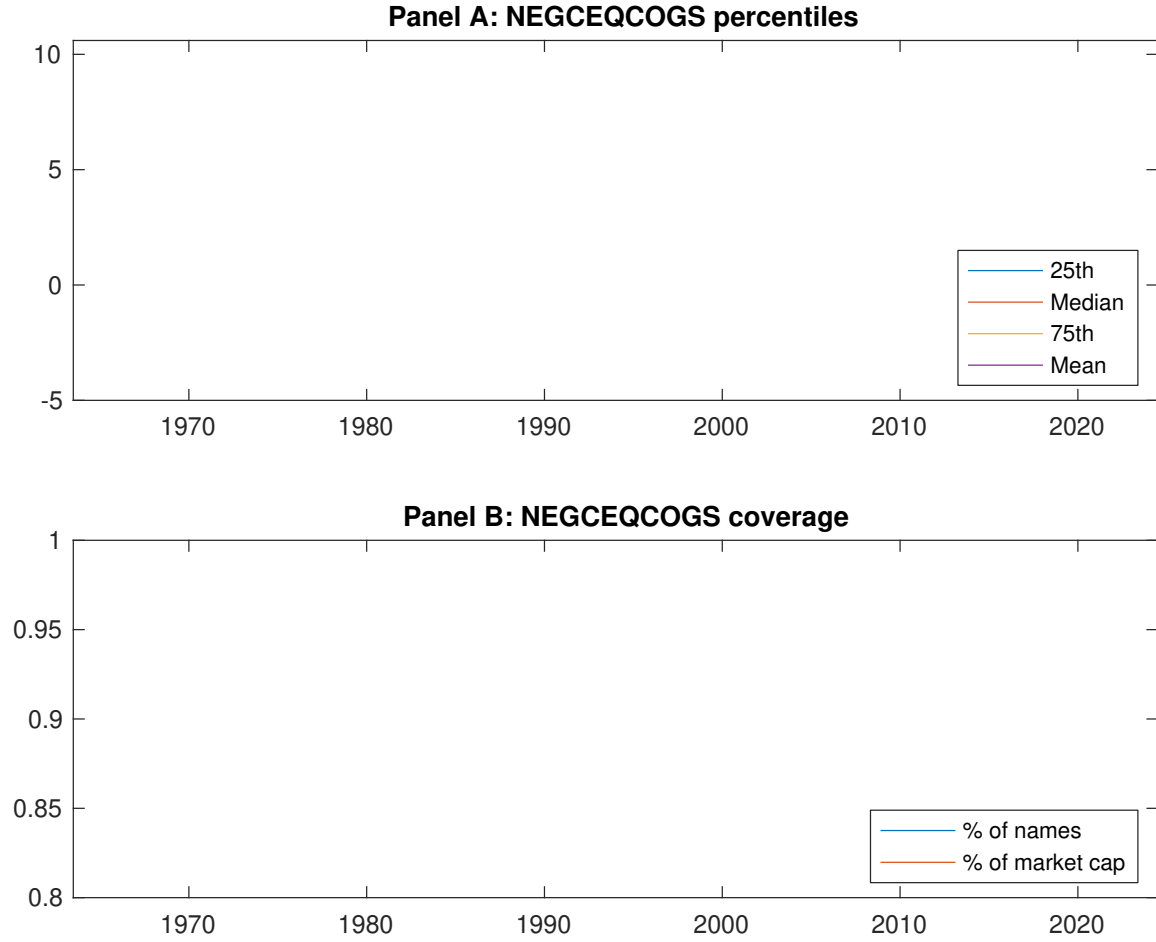


Figure 1: Times series of EPI percentiles and coverage.
This figure plots descriptive statistics for EPI. Panel A shows cross-sectional percentiles of EPI over the sample. Panel B plots the monthly coverage of EPI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EPI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EPI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [3.27]	0.56 [3.43]	0.62 [3.56]	0.67 [3.74]	0.72 [3.91]	0.19 [1.94]
α_{CAPM}	-0.03 [-0.57]	-0.00 [-0.11]	0.02 [0.48]	0.07 [1.13]	0.11 [1.63]	0.13 [1.37]
α_{FF3}	0.01 [0.27]	-0.02 [-0.40]	-0.01 [-0.24]	0.03 [0.48]	0.06 [0.95]	0.05 [0.52]
α_{FF4}	0.03 [0.71]	-0.01 [-0.35]	0.01 [0.26]	0.04 [0.63]	0.05 [0.73]	0.02 [0.17]
α_{FF5}	0.12 [3.20]	-0.05 [-1.30]	-0.11 [-2.39]	-0.08 [-1.37]	-0.05 [-0.80]	-0.17 [-2.09]
α_{FF6}	0.13 [3.32]	-0.05 [-1.15]	-0.08 [-1.76]	-0.06 [-1.04]	-0.05 [-0.85]	-0.18 [-2.18]
Panel B: Fama and French (2018) 6-factor model loadings for EPI-sorted portfolios						
β_{MKT}	0.94 [100.07]	1.00 [106.05]	1.04 [95.73]	1.04 [76.11]	1.04 [73.89]	0.10 [5.13]
β_{SMB}	-0.17 [-12.72]	-0.09 [-6.57]	0.08 [4.86]	0.11 [5.82]	0.27 [13.10]	0.44 [15.38]
β_{HML}	0.01 [0.58]	-0.00 [-0.20]	0.03 [1.39]	0.00 [0.19]	0.07 [2.70]	0.06 [1.65]
β_{RMW}	-0.24 [-12.92]	0.03 [1.58]	0.25 [11.69]	0.24 [8.85]	0.32 [11.67]	0.56 [14.45]
β_{CMA}	-0.17 [-6.36]	0.13 [4.83]	0.09 [2.86]	0.15 [3.86]	-0.01 [-0.27]	0.16 [2.83]
β_{UMD}	-0.01 [-0.98]	-0.01 [-0.80]	-0.04 [-3.72]	-0.03 [-1.87]	0.01 [0.37]	0.01 [0.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1140	676	595	597	714	
me (\$10 ⁶)	3543	2221	1557	1651	1321	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EPI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19 [1.94]	0.13 [1.37]	0.05 [0.52]	0.02 [0.17]	-0.17 [-2.09]	-0.18 [-2.18]
Quintile	NYSE	EW	0.19 [1.96]	0.10 [1.04]	-0.00 [-0.01]	0.04 [0.42]	-0.13 [-1.59]	-0.09 [-1.05]
Quintile	Name	VW	0.30 [2.76]	0.25 [2.34]	0.17 [1.64]	0.11 [1.07]	-0.12 [-1.33]	-0.14 [-1.60]
Quintile	Cap	VW	0.22 [2.20]	0.19 [1.83]	0.10 [1.00]	0.05 [0.51]	-0.19 [-2.15]	-0.20 [-2.30]
Decile	NYSE	VW	0.26 [2.32]	0.19 [1.72]	0.11 [1.05]	0.05 [0.47]	-0.14 [-1.47]	-0.17 [-1.74]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [1.77]	0.11 [1.16]	0.04 [0.46]	0.02 [0.25]	0.12 [1.42]	0.12 [1.50]
Quintile	NYSE	EW	0.06 [0.65]					
Quintile	Name	VW	0.28 [2.59]	0.23 [2.11]	0.16 [1.54]	0.12 [1.20]	0.05 [0.56]	0.07 [0.75]
Quintile	Cap	VW	0.21 [2.06]	0.16 [1.59]	0.09 [0.89]	0.06 [0.60]	0.13 [1.45]	0.14 [1.58]
Decile	NYSE	VW	0.24 [2.12]	0.17 [1.49]	0.10 [0.96]	0.06 [0.62]	0.06 [0.67]	0.08 [0.86]

Table 3: Conditional sort on size and EPI

This table presents results for conditional double sorts on size and EPI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EPI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EPI and short stocks with low EPI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EPI Quintiles					EPI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.21]	0.76 [3.17]	0.81 [3.33]	0.83 [3.39]	0.87 [3.15]	0.36 [2.64]	0.26 [1.95]	0.18 [1.33]	0.23 [1.72]	0.05 [0.39]	0.11 [0.84]
	(2)	0.58 [2.57]	0.86 [3.91]	0.83 [3.71]	0.74 [3.25]	0.89 [3.64]	0.31 [2.56]	0.23 [1.96]	0.13 [1.10]	0.18 [1.48]	-0.17 [-1.57]	-0.10 [-0.94]
	(3)	0.51 [2.42]	0.67 [3.40]	0.77 [3.78]	0.82 [3.89]	0.93 [4.12]	0.42 [3.32]	0.35 [2.81]	0.23 [1.86]	0.23 [1.85]	-0.12 [-1.10]	-0.09 [-0.80]
	(4)	0.62 [3.00]	0.63 [3.50]	0.75 [3.95]	0.72 [3.56]	0.75 [3.67]	0.13 [0.99]	0.12 [0.89]	0.00 [0.01]	-0.01 [-0.05]	-0.37 [-3.46]	-0.35 [-3.21]
	(5)	0.47 [2.82]	0.52 [3.16]	0.53 [3.26]	0.64 [3.81]	0.62 [3.53]	0.15 [1.42]	0.11 [1.06]	0.04 [0.40]	-0.01 [-0.09]	-0.18 [-1.91]	-0.21 [-2.14]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EPI Quintiles					EPI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	418	418	418	416	402	36	36	35	35	31	
	(2)	116	116	116	116	116	58	60	60	59	59	
	(3)	83	83	83	83	83	101	102	103	104	104	
	(4)	68	69	69	69	69	214	217	221	216	220	
(5)	62	62	63	62	62	1861	1832	1438	1670	1421		

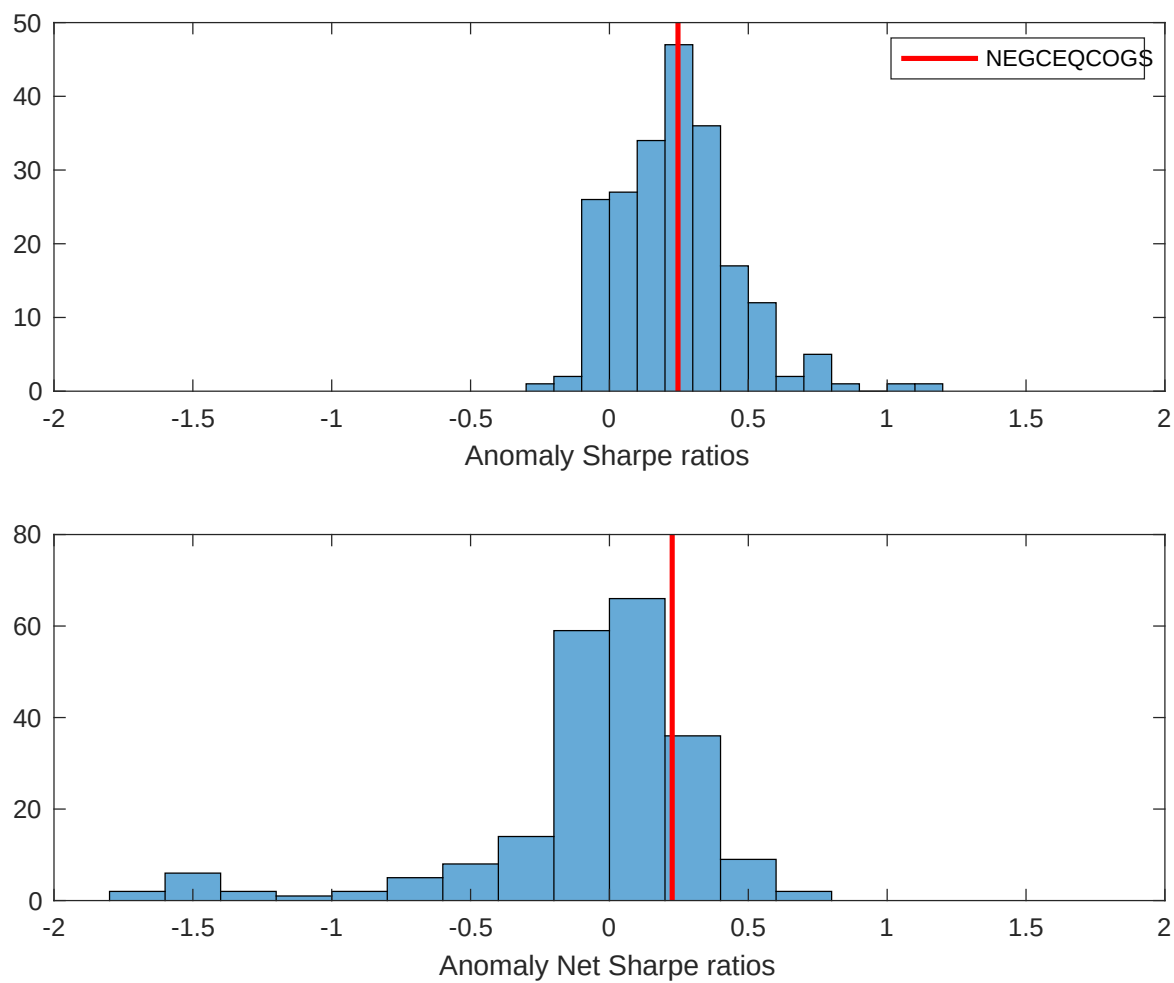


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EPI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

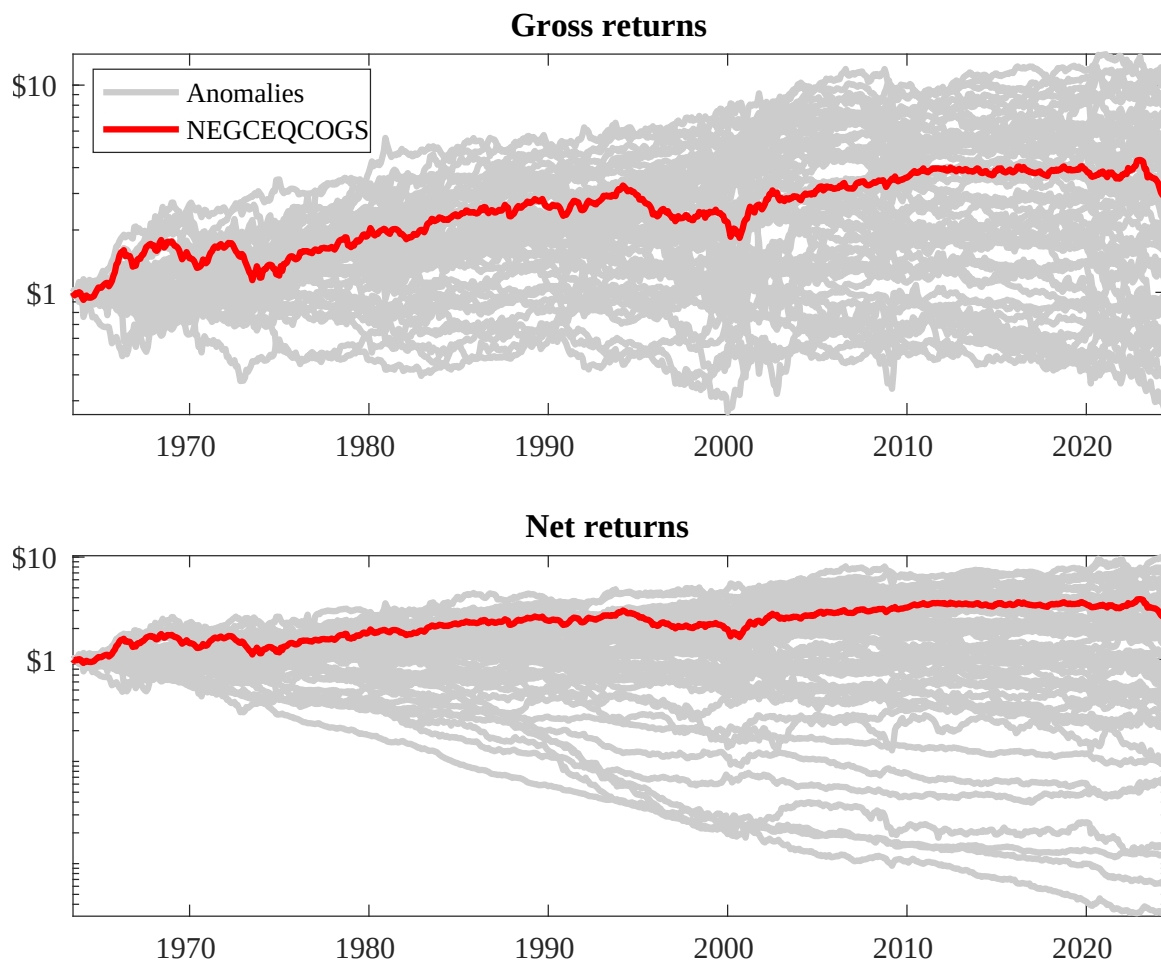
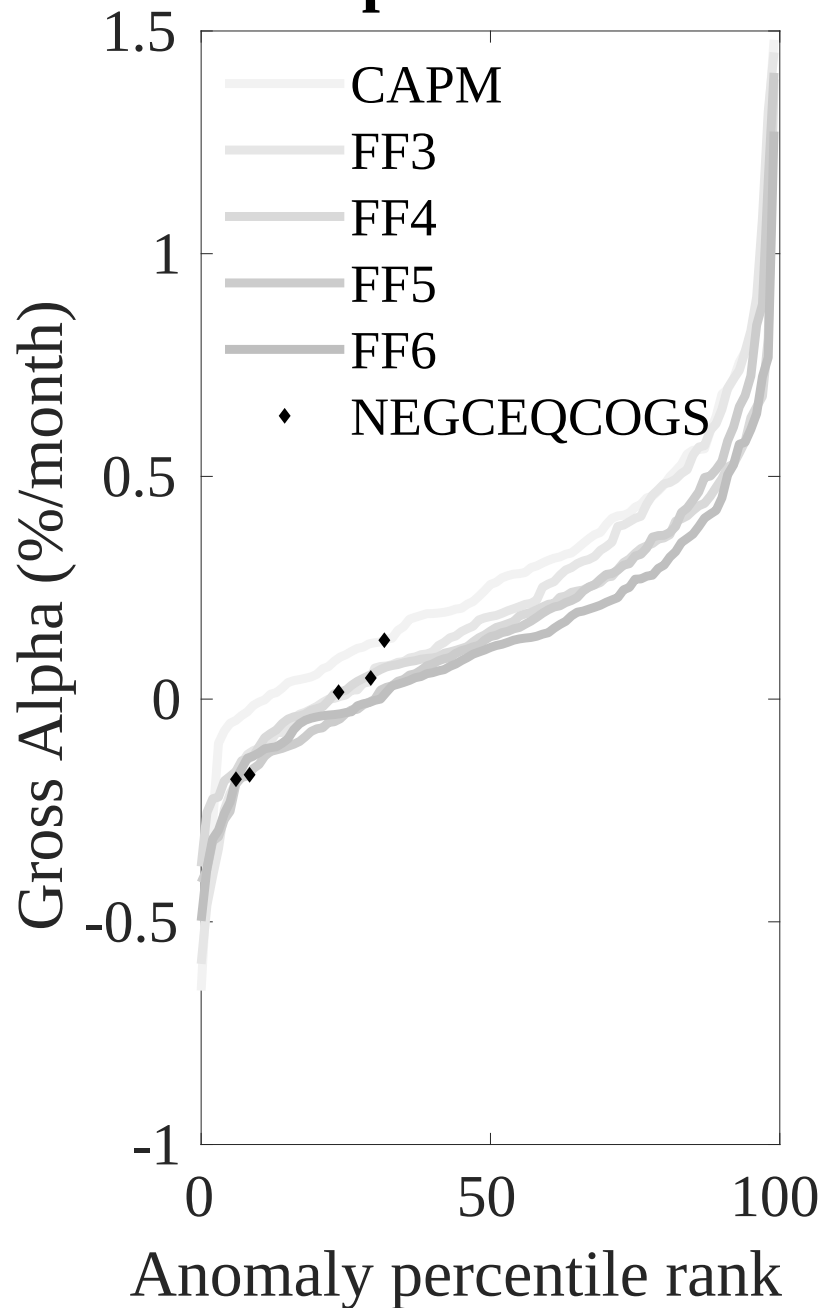


Figure 3: Dollar invested.

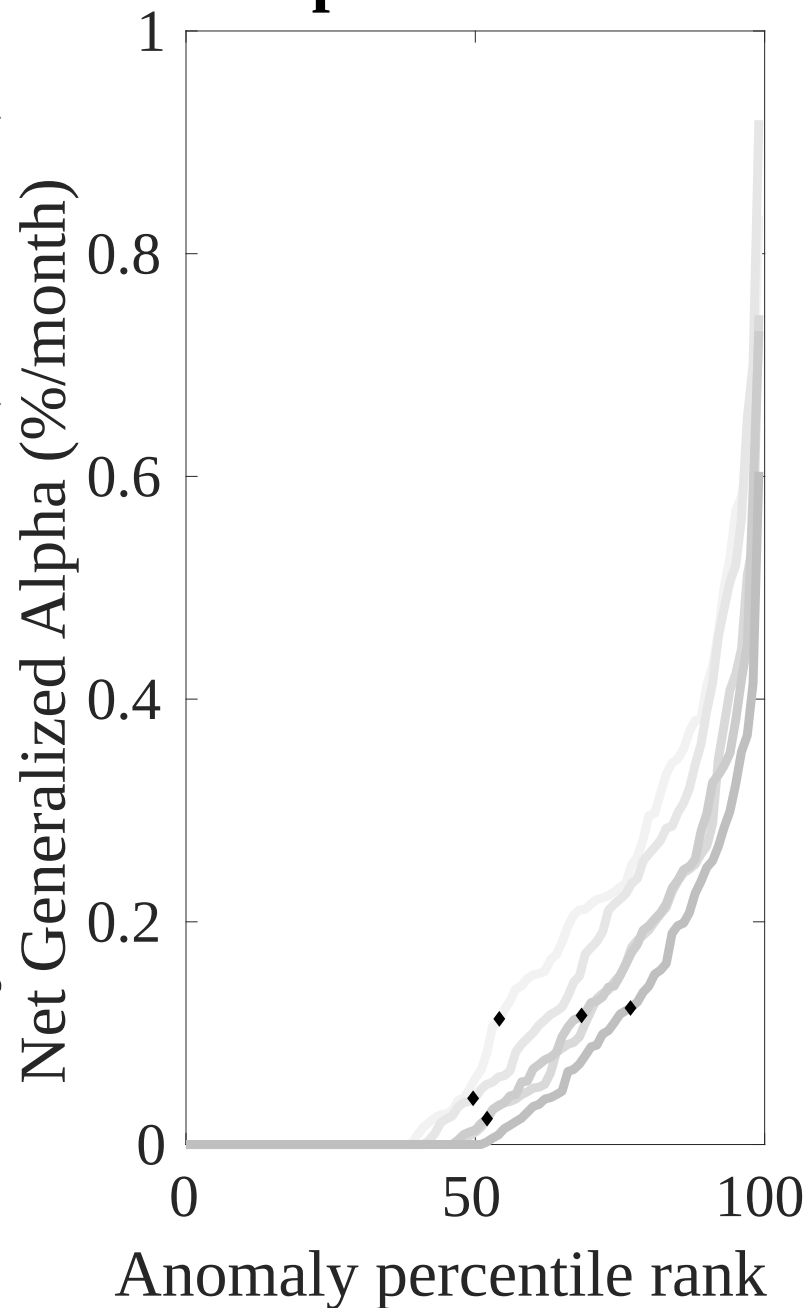
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EPI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



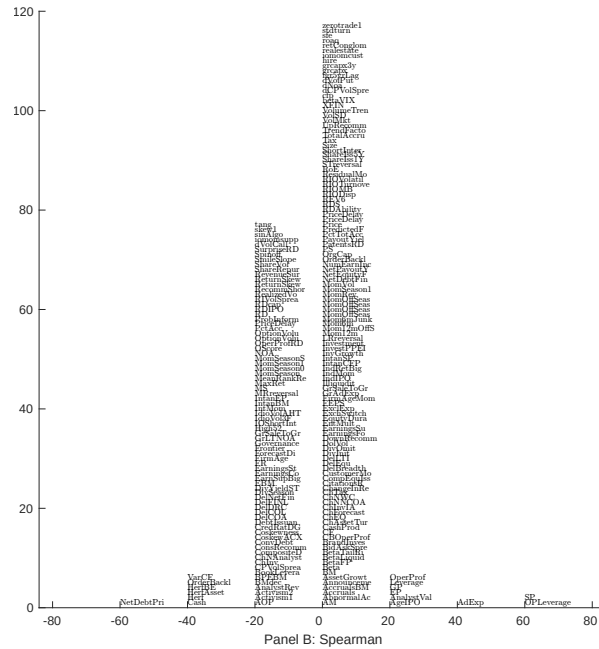
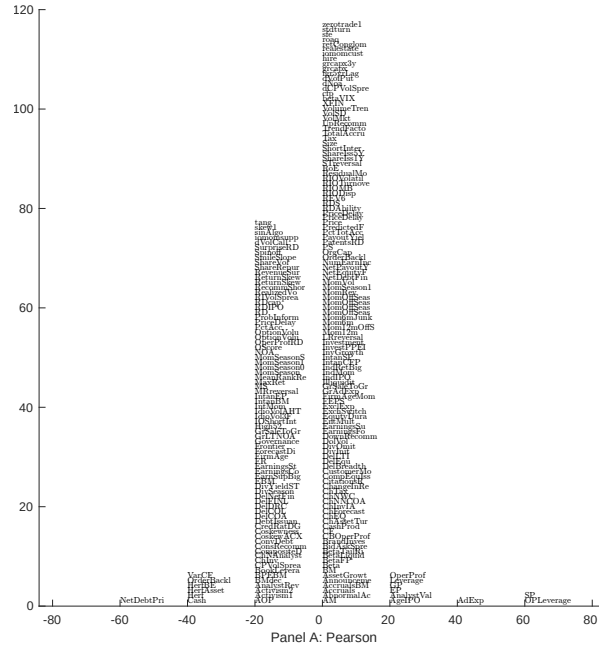


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EPI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

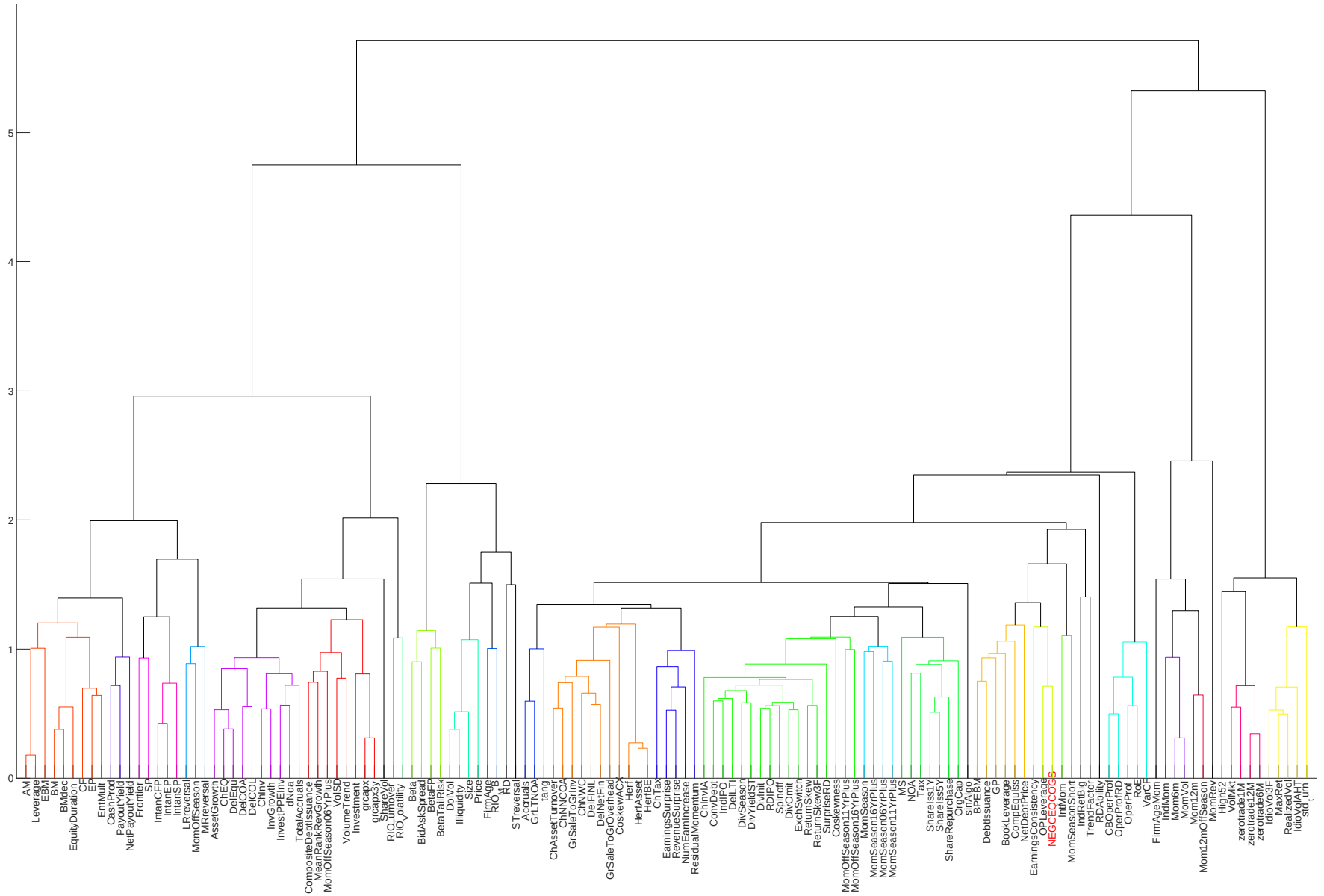


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

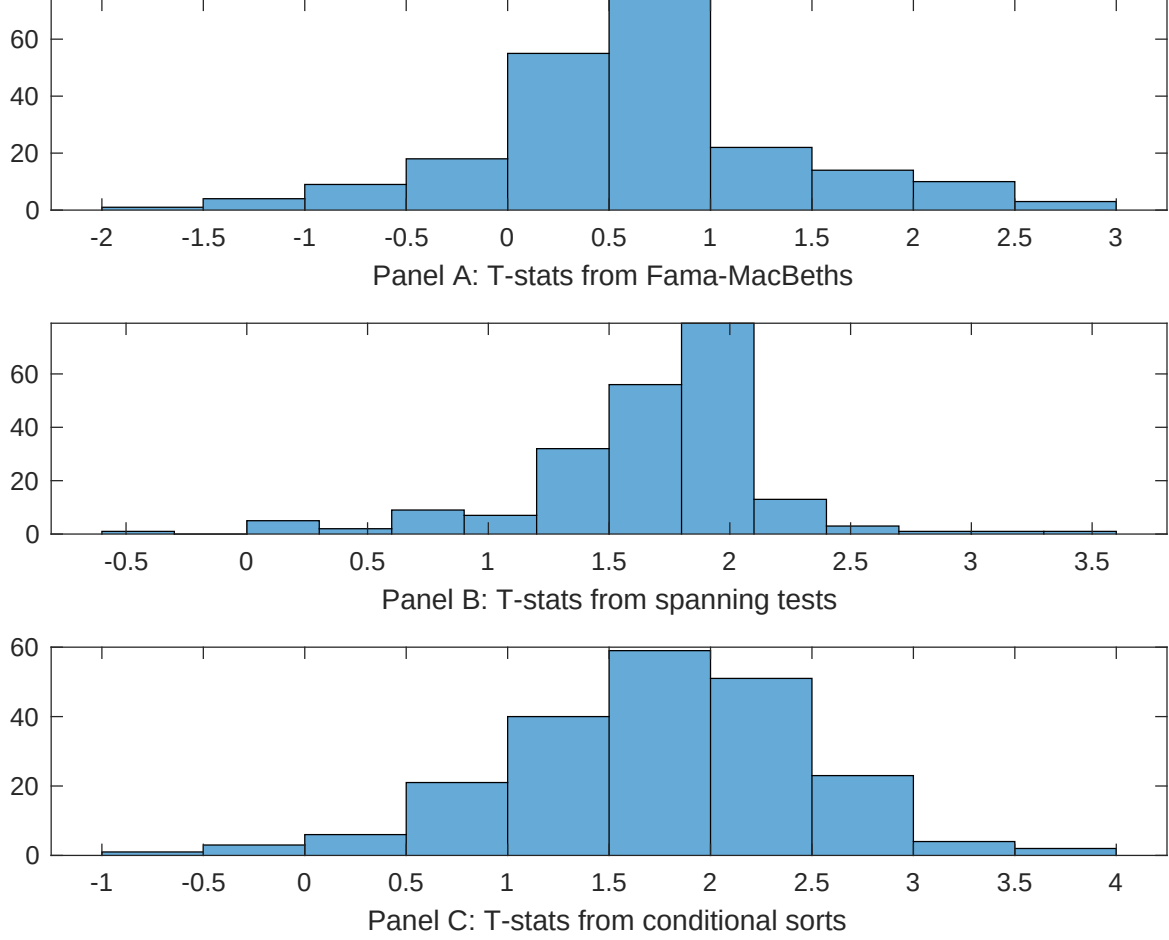


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EPI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.10 [5.00]	0.98 [4.75]	0.12 [5.74]	0.11 [4.69]	0.14 [5.14]	0.12 [5.01]	0.76 [2.33]
EPI	-0.59 [-0.53]	-0.13 [-1.22]	0.53 [0.41]	-0.16 [-1.13]	0.76 [2.52]	0.45 [2.40]	-0.54 [-0.66]
Anomaly 1	0.13 [2.88]						0.87 [0.58]
Anomaly 2		0.69 [4.09]					0.96 [2.27]
Anomaly 3			-0.37 [-1.88]				0.12 [1.54]
Anomaly 4				0.26 [0.52]			-0.69 [-0.53]
Anomaly 5					0.71 [0.02]		0.17 [0.11]
Anomaly 6						0.87 [2.47]	0.19 [4.16]
# months	732	727	727	619	636	626	582
$\bar{R}^2(\%)$	0	1	1	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EPI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EPI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.27 [-4.00]	-0.25 [-3.74]	-0.16 [-1.98]	-0.20 [-2.38]	-0.17 [-2.00]	-0.10 [-1.15]	-0.21 [-3.56]
Anomaly 1	53.43 [19.89]						43.85 [17.69]
Anomaly 2		53.27 [18.82]					27.00 [9.03]
Anomaly 3			-22.47 [-7.30]				-3.66 [-1.47]
Anomaly 4				11.11 [5.19]			4.32 [2.34]
Anomaly 5					-12.12 [-6.16]		-10.17 [-7.40]
Anomaly 6						-20.36 [-6.26]	-17.42 [-7.56]
mkt	10.58 [6.62]	3.30 [1.98]	3.89 [1.85]	9.90 [5.06]	9.46 [4.81]	12.66 [6.22]	8.82 [5.87]
smb	20.11 [7.72]	10.86 [3.69]	27.86 [7.82]	39.82 [13.53]	35.43 [11.89]	37.99 [12.90]	-3.42 [-1.25]
hml	34.86 [10.26]	-25.83 [-7.18]	-2.99 [-0.76]	1.62 [0.42]	0.70 [0.19]	-4.33 [-1.10]	1.22 [0.36]
rmw	26.56 [7.70]	41.61 [12.74]	63.17 [16.34]	54.03 [14.10]	48.01 [12.14]	45.52 [11.17]	11.12 [3.42]
cma	5.16 [1.13]	1.54 [0.33]	11.05 [2.02]	10.52 [1.86]	25.01 [4.36]	16.26 [2.78]	-4.49 [-1.11]
umd	0.30 [0.19]	22.95 [11.61]	4.57 [2.35]	5.39 [2.59]	-2.31 [-1.18]	-0.20 [-0.10]	12.22 [7.16]
# months	732	728	728	716	642	629	629
$\bar{R}^2(\%)$	58	57	40	38	37	38	72

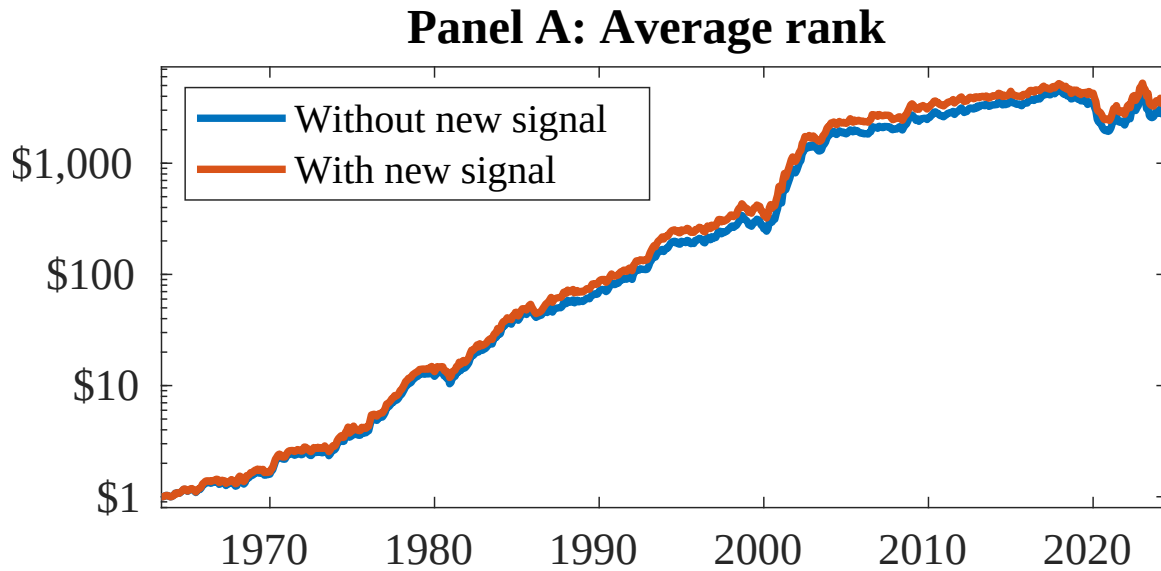


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EPI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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